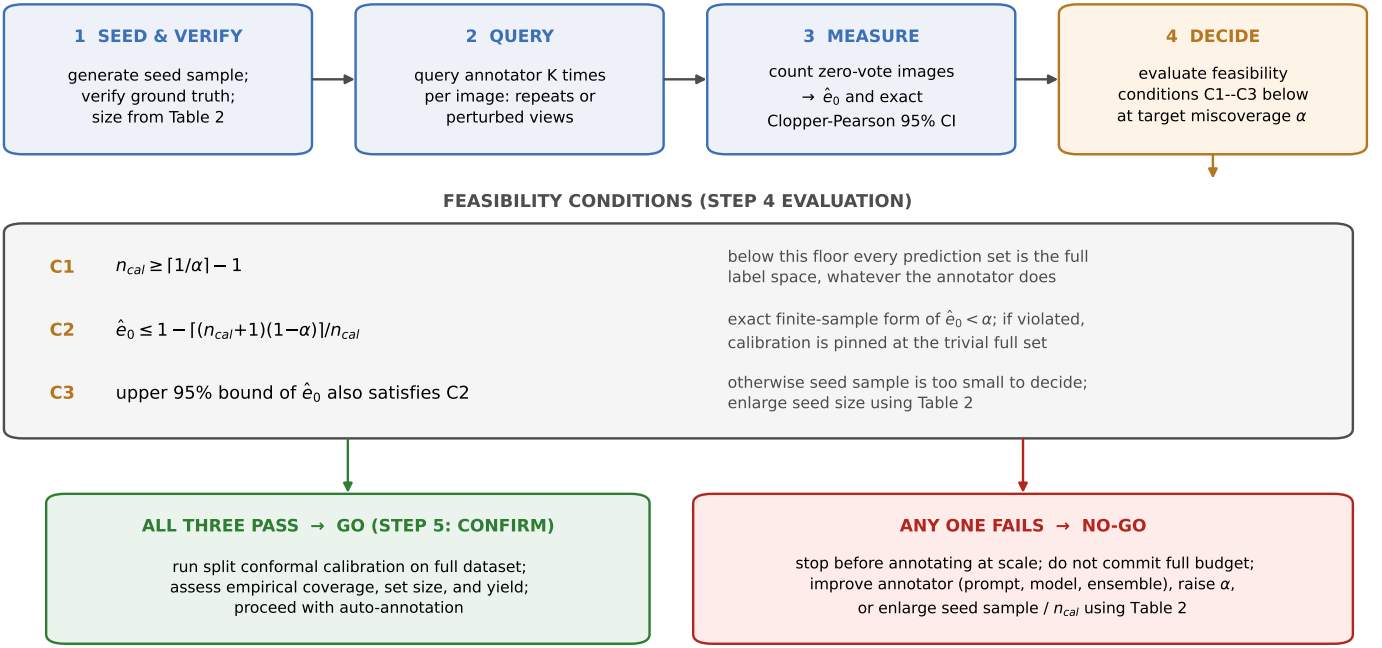
1 SEED & VERIFY
generate seed sample; verify ground truth; size from Table 2
2 QUERY
query annotator K times per image: repeats or perturbed views
3 MEASURE
count zero-vote images → $\hat{e}_0$ and exact Clopper-Pearson 95% CI
4 DECIDE
evaluate feasibility conditions C1--C3 below at target miscoverage α
FEASIBILITY CONDITIONS (STEP 4 EVALUATION)
C1
$n_{cal} \geq \lceil 1/\alpha \rceil - 1$
below this floor every prediction set is the full label space, whatever the annotator does
C2
$\hat{e}_0 \leq 1 - \lceil (n_{cal}+1)(1-\alpha) \rceil / n_{cal}$
exact finite-sample form of $\hat{e}_0 < \alpha$; if violated, calibration is pinned at the trivial full set
C3
upper 95% bound of $\hat{e}_0$ also satisfies C2
otherwise seed sample is too small to decide; enlarge seed size using Table 2
ALL THREE PASS → GO (STEP 5: CONFIRM)
run split conformal calibration on full dataset; assess empirical coverage, set size, and yield; proceed with auto-annotation
ANY ONE FAILS → NO-GO
stop before annotating at scale; do not commit full budget; improve annotator (prompt, model, ensemble), raise α, or enlarge seed sample / n_{cal} using Table 2